

ERGA Assembly Report

v24.10.15

Tags: ERGA-Community

TxID	880251
ToLID	fZinAsp1
Species	Zingel asper
Class	Actinopteri
Order	Perciformes

Genome Traits	Expected	Observed
Haploid size (bp)	843,563,058	963,590,568
Haploid Number	24 (source: ancestor)	24
Ploidy	2 (source: ancestor)	2
Sample Sex	male	unknown

EBP metrics summary and curation notes

Obtained EBP quality metric for pri: 6.7.Q55

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

- . Observed sex is different from Sample sex
- . Assembly length loss > 3% for pri

Curator notes

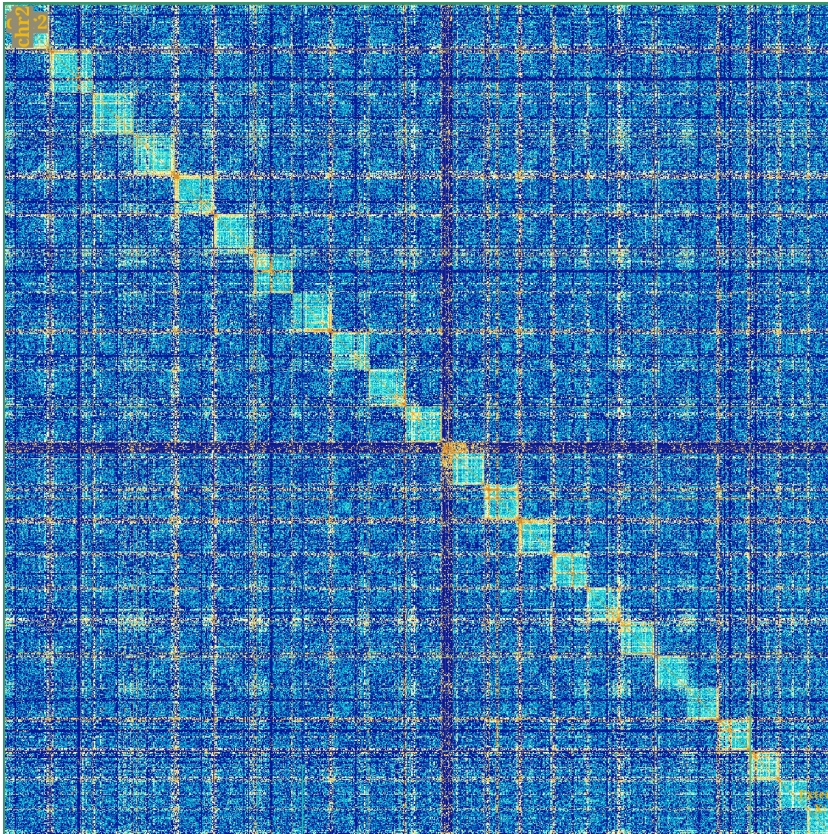
. Interventions/Gb: 0
. Contamination notes: "Blobtools detected presence of bacteria contaminants, which were removed. Mitochondrial genome was removed from the assembly"
. Other observations: "Sex chromosomes not found, Zingel zingel, x1x2y - x1x2, GC content scaffolds that were not between 0,38 and 0,463, were smaller than 14400bp and had less coverage than 6,4x were filtered out through blobtools. Also, those sequences whose blast results were assigned to other lineages than Chordata were spurious and therefore, removed. Manual curation using Hi-C contact maps resulted in 21 joins and rearrangement of the assembly. The assembly contained 293 scaffolds before manual curation and 267 scaffolds in the final curated assembly."

Quality metrics table

Metrics	Pre-curation pri	Curated pri
Total bp	1,014,352,583	963,590,568
GC %	41.09	40.86
Gaps/Gbp	0	752.39
Total gap bp	0	142,900
Scaffolds	1,747	267
Scaffold N50	3,849,056	38,905,730
Scaffold L50	68	12
Scaffold L90	310	23
Contigs	1,747	992
Contig N50	3,849,056	3,215,671
Contig L50	68	88
Contig L90	310	327
QV	55.6548	55.8612
Kmer compl.	95.7086	95.3399
BUSCO sing.	97.8%	98.4%
BUSCO dupl.	0.9%	0.7%
BUSCO frag.	0.7%	0.2%
BUSCO miss.	0.6%	0.7%

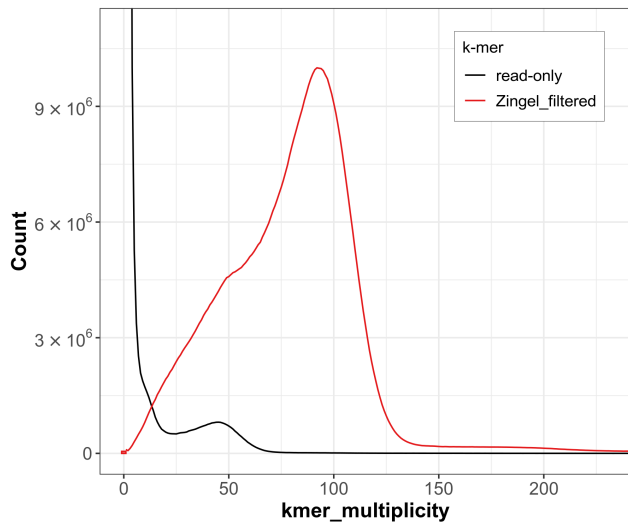
BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: actinopterygii_odb12 (genomes:75, BUSCOs:7207)

HiC contact map of curated assembly

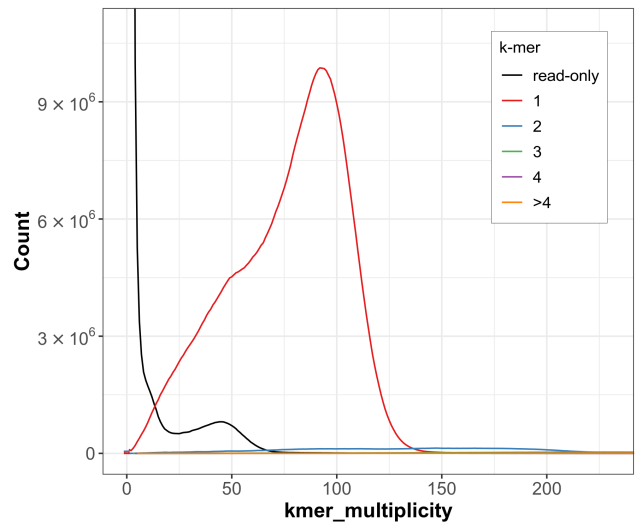


pr*i* [\[LINK\]](#)

K-mer spectra of curated assembly

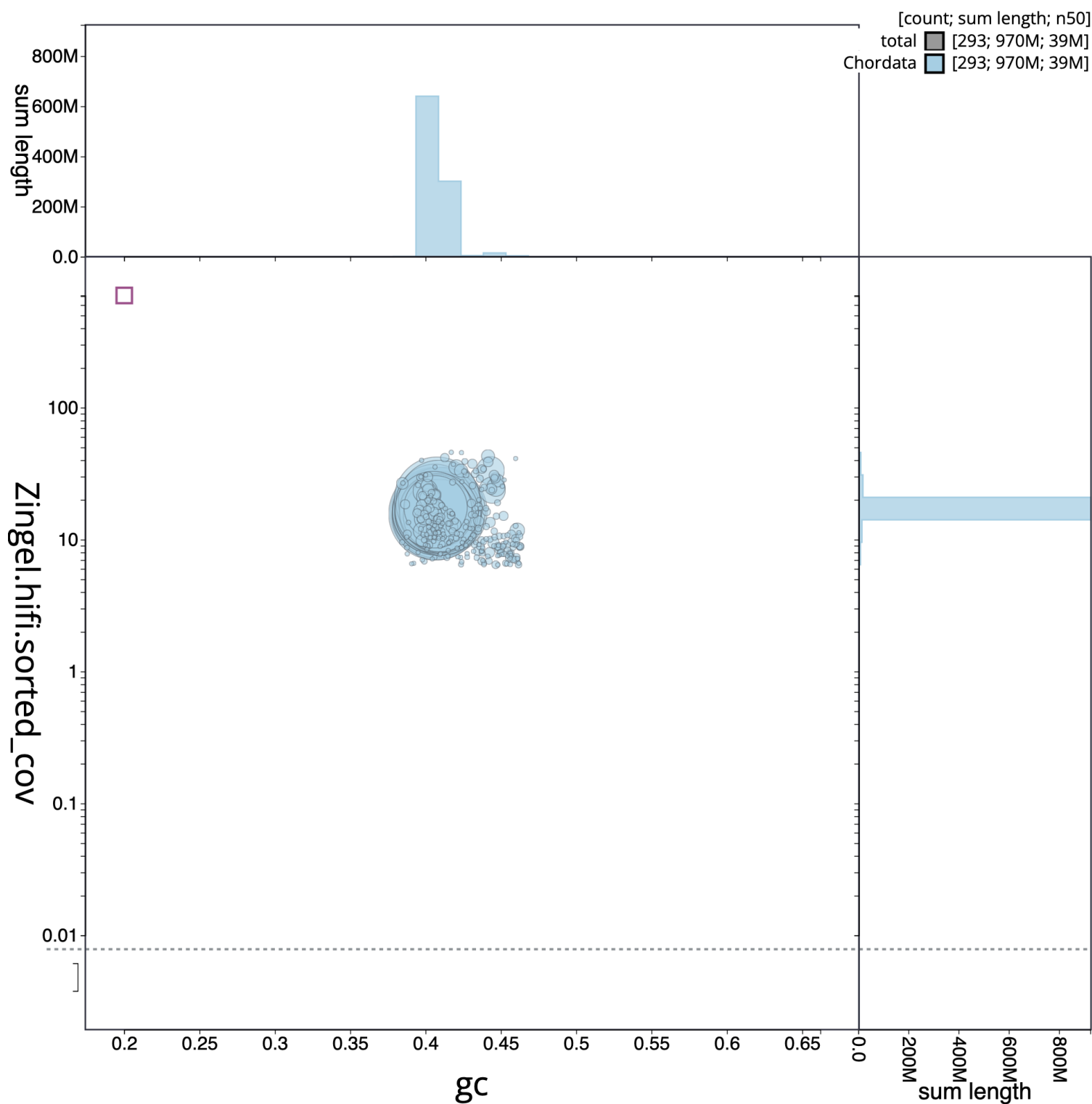


Distribution of k-mer counts coloured by their presence in reads/assemblies



Distribution of k-mer counts per copy numbers found in asm

Post-curation contamination screening



pri. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	HiFi	Bionano	HiC
Coverage	40x	NA	60x

Assembly pipeline

- **Hifiasm**
 - |_ *ver*: 0.25.40
 - |_ *key param*: primary
- **purge_dups**
 - |_ *ver*: 1.2.5
 - |_ *key param*: NA
- **YaHS**
 - |_ *ver*: 1.2.2
 - |_ *key param*: NA

Curation pipeline

- **BlobtoolKit**
 - |_ *ver*: 2.1.1
 - |_ *key param*: NA

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